CSET485 — ASSIGNMENT 1

STATISTICAL ANALYSIS, MODEL EVALUATION AND ALGORITHMIC FAIRNESS

Dataset : COMPAS Recidivism Dataset

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Course Code : CSET485

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1. Introduction

In this assignment, I worked with the COMPAS recidivism dataset to understand how statistical models make predictions and how fair those predictions really are. The main goal was to see how different models behave, how the results change when we look at different demographic groups, and how the assumptions we make can affect the conclusions we draw.

First, I compared the classifications from linear regression and logistic regression. Then I ran the regression separately for different subgroups to see whether the results stayed the same. I also looked at omitted-variable bias, which shows how leaving out an important variable can change the results. After that, I tested a selected set of 50 adversarial cases to see where the model struggles. In the last part, I checked fairness using three measures: demographic parity, equalized odds, and predictive parity, at different decision thresholds.

One thing I learned is that accuracy alone is not enough to judge a model. Things like how the data is preprocessed, which threshold is chosen, how the model performs for each subgroup, and what happens when it predicts wrongly all matter a lot. Since this is observational data, I have also tried to keep in mind the uncertainty in the results and the possibility of confounding.

Overall, this assignment helped me understand the strengths and limitations of these models, and it made me think about how much evidence we would need before using recidivism predictions to make serious real-world decisions.

2. Dataset and Methodology

2.1 Dataset Overview

For this assignment, I used the COMPAS recidivism dataset to study how statistical models predict whether a person will reoffend within two years. The dataset has demographic details, criminal history variables, and the final recidivism outcome. Using these, I could check how things like prior records, age, and demographic groups are related to the predicted outcome.

2.2 Model Comparison

In the first part, I compared Linear Regression and Logistic Regression. I took 2,165 test cases and classified them using the thresholds given in the notebook. Then I checked how often the two models agreed and where they gave different results.

2.3 Subgroup Regression

Here I fitted logistic regression on the full dataset and then separately on the African-American and Caucasian subgroups. I used prior criminal-record count and age as the predictors for two-year recidivism. To understand the results, I looked at the coefficients, confidence intervals, and p-values.

2.4 Omitted-Variable Bias

In this experiment, I removed some predictors from the regression models to see how the estimated relationships changed. The variables I worked with were age, juvenile felony count, juvenile misdemeanor count, and juvenile other-offence count.

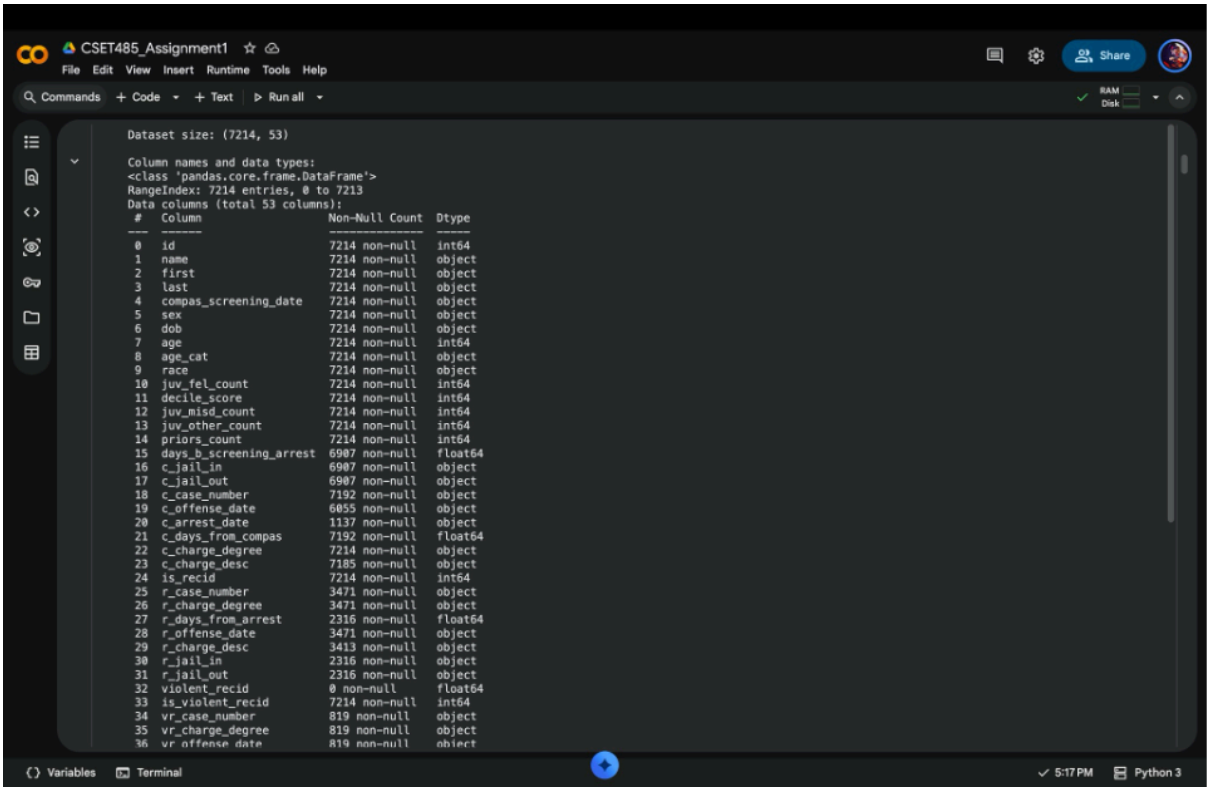
2.5 Adversarial Subset Analysis

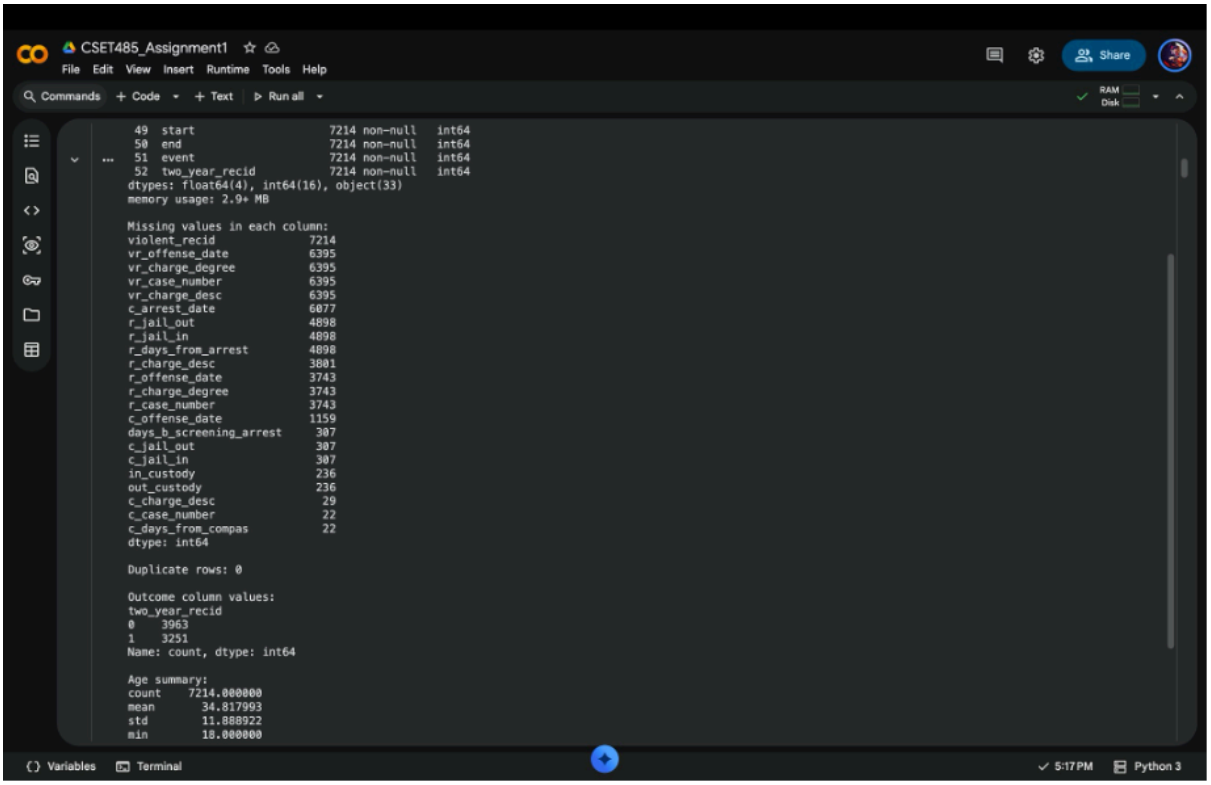
For this part, I picked a subset of 50 cases and studied its characteristics, mainly the prior criminal-record counts. Since the sample is small and was specially selected, I was careful while interpreting it because it may not represent the full dataset.

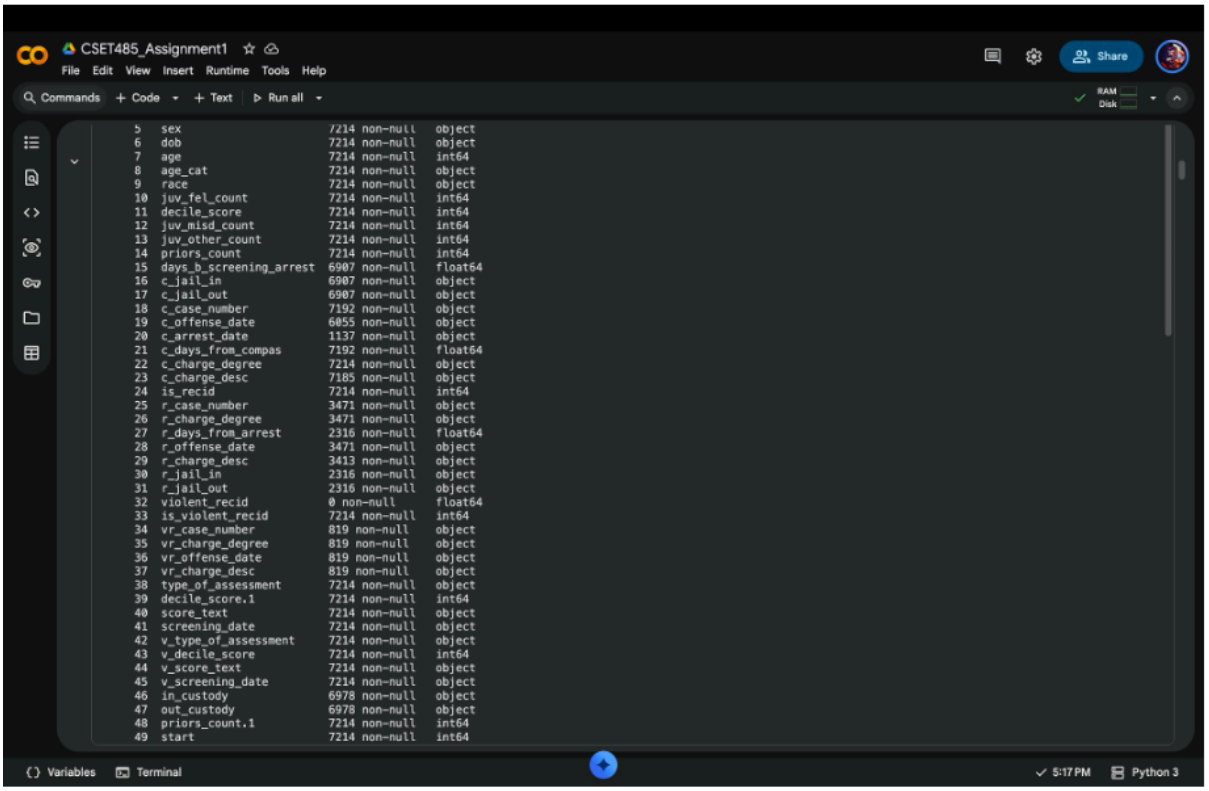
2.6 Fairness Evaluation

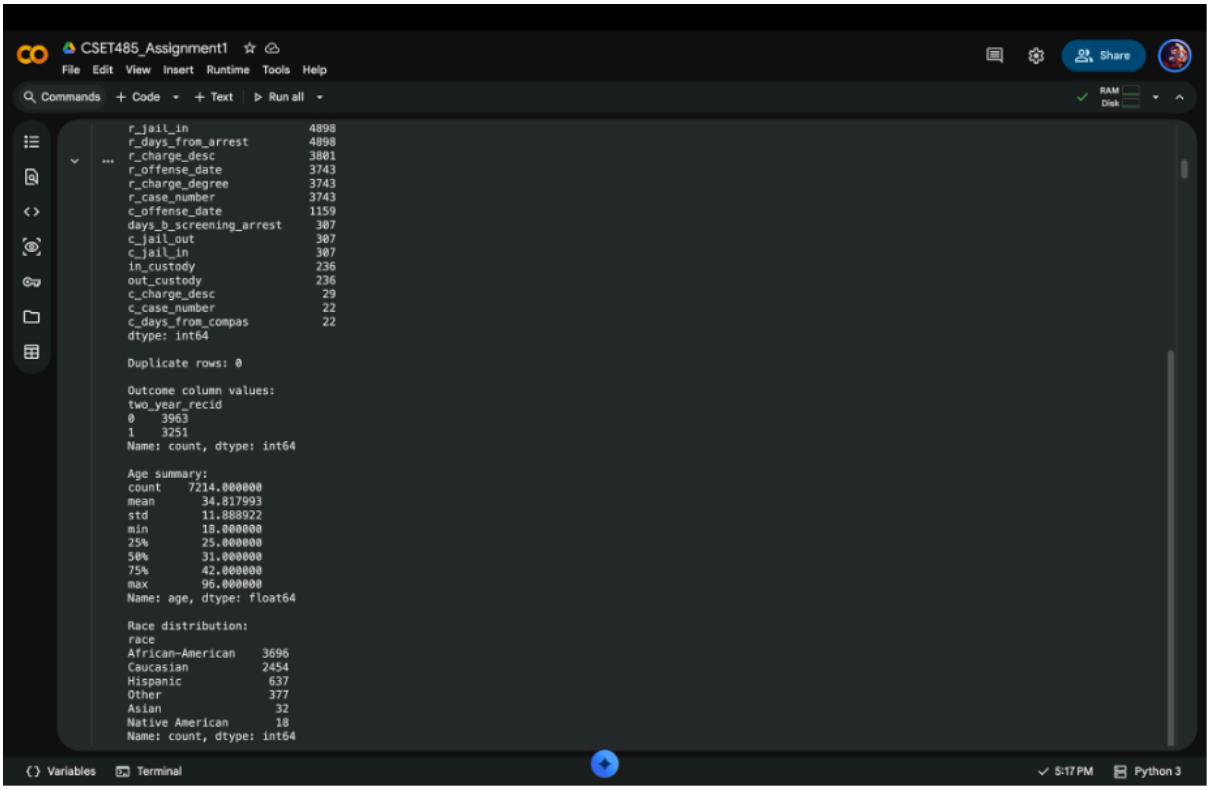
In the last analysis, I compared demographic parity, equalized odds, and predictive parity at decision thresholds from 0.05 to 0.95. The aim was to see how changing the threshold affects the fairness measures and what trade-offs come up when a model is actually used.

Screenshots :-




CSET485_Assignment1
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
49 start 7214 non-null int64
50 end 7214 non-null int64
51 event 7214 non-null int64
52 two_year_recid 7214 non-null int64
dtypes: float64(4), int64(16), object(33)
memory usage: 2.9+ MB
Missing values in each column:
violent_recid 7214
vr_offense_date 6395
vr_charge_degree 6395
vr_case_number 6395
vr_charge_desc 6395
c_arrest_date 6077
r_jail_out 4898
r_jail_in 4898
r_days_from_arrest 4898
r_charge_desc 3801
r_offense_date 3743
r_charge_degree 3743
r_case_number 3743
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days_b_screening_arrest 307
c_jail_out 307
c_jail_in 307
in_custody 236
out_custody 236
c_charge_desc 29
c_case_number 22
c_days_from_compas 22
dtype: int64
Duplicate rows: 0
Outcome column values:
two_year_recid
0 3963
1 3251
Name: count, dtype: int64
Age summary:
count 7214.000000
mean 34.817993
std 11.888922
min 18.000000
Variables Terminal
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CSET485_Assignment1
File Edit View Insert Runtime Tools Help
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Commands + Code + Text Run all
RAM Disk
5 sex 7214 non-null object
6 dob 7214 non-null object
7 age 7214 non-null int64
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9 race 7214 non-null object
10 juv_fel_count 7214 non-null int64
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13 juv_other_count 7214 non-null int64
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19 c_offense_date 6055 non-null object
20 c_arrest_date 1137 non-null object
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22 c_charge_degree 7214 non-null object
23 c_charge_desc 7185 non-null object
24 is_recid 7214 non-null int64
25 r_case_number 3471 non-null object
26 r_charge_degree 3471 non-null object
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31 r_jail_out 2316 non-null object
32 violent_recid 0 non-null float64
33 is_violent_recid 7214 non-null int64
34 vr_case_number 819 non-null object
35 vr_charge_degree 819 non-null object
36 vr_offense_date 819 non-null object
37 vr_charge_desc 819 non-null object
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43 v_decile_score 7214 non-null int64
44 v_score_text 7214 non-null object
45 v_screening_date 7214 non-null object
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47 out_custody 6978 non-null object
48 priors_count.1 7214 non-null int64
49 start 7214 non-null int64
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3. Results and Discussion

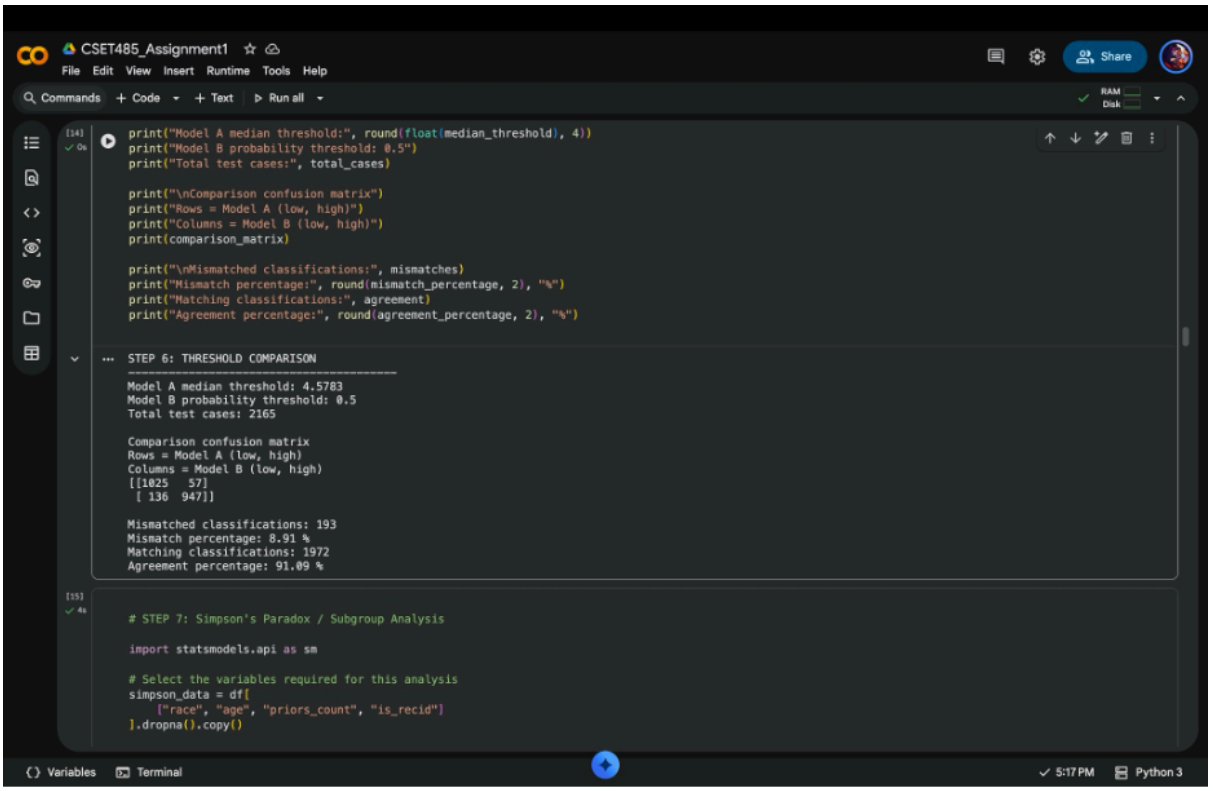
3.1 Comparison of the Two Models

I compared Linear Regression and Logistic Regression on 2,165 test cases. Both models gave the same classification for 1,972 cases, which is 91.09% agreement. The remaining 193 cases (8.91%) were classified differently.

From the comparison matrix, 57 cases were low according to Model A but high according to Model B. The other 136 cases were high according to Model A but low according to Model B. This shows that the choice of model and threshold can change the prediction for an individual person.

Linear Regression predicts a numerical value, while Logistic Regression gives a probability for a yes/no outcome. So their scales and thresholds don't mean the same thing. I think these mismatched cases should be studied properly before anyone uses such predictions for important decisions.

Screenshot :-



Model Comparison Output and Confusion Matrix

The output compares the classifications produced by Linear Regression and Logistic Regression on 2,165 test cases. The models agreed on 1,972 cases (91.09%) and disagreed on 193 cases (8.91%). The comparison confusion matrix shows the distribution of matching and mismatching classifications.

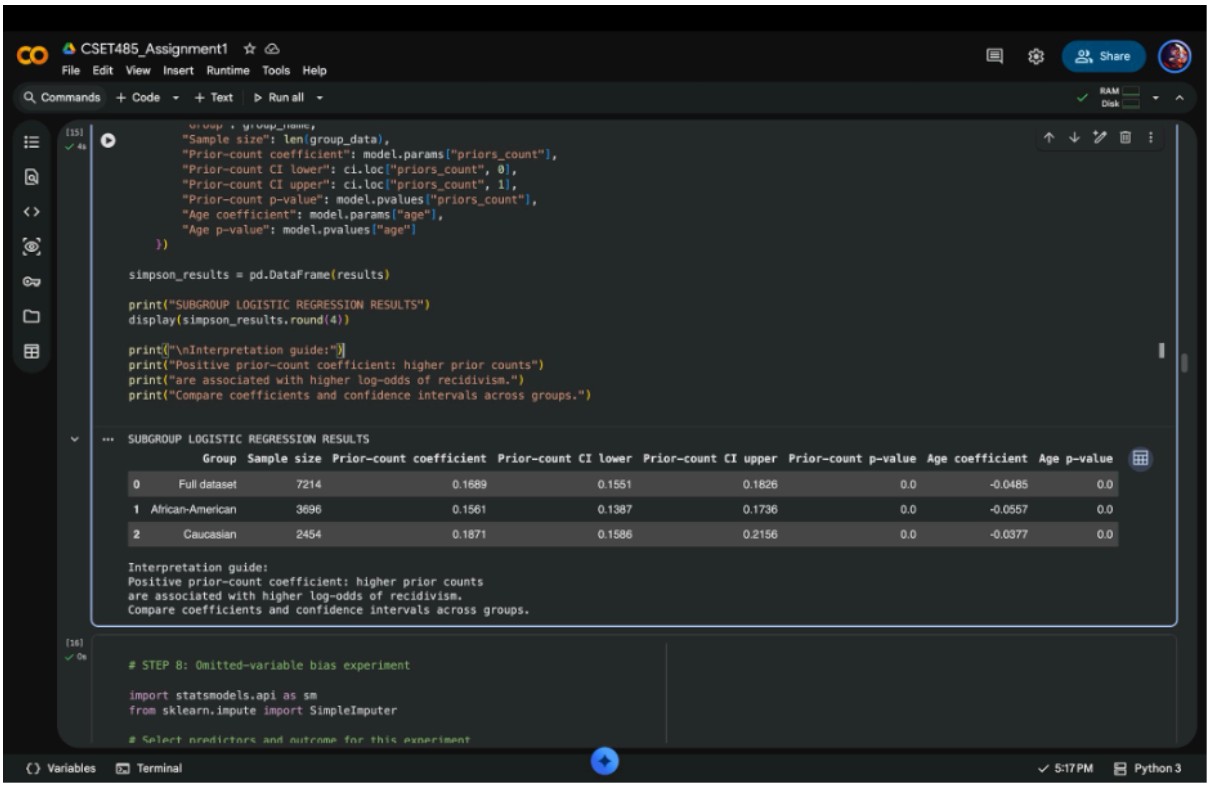
3.2 Subgroup Regression Results

The full dataset had 7,214 observations. The African-American subgroup had 3,696 observations and the Caucasian subgroup had 2,454.

The prior-count coefficient was 0.1689 for the full dataset (95% CI: [0.1551, 0.1826]). For the African-American subgroup it was 0.1561 (CI: [0.1387, 0.1736]), and for the Caucasian subgroup it was 0.1871 (CI: [0.1586, 0.2156]).

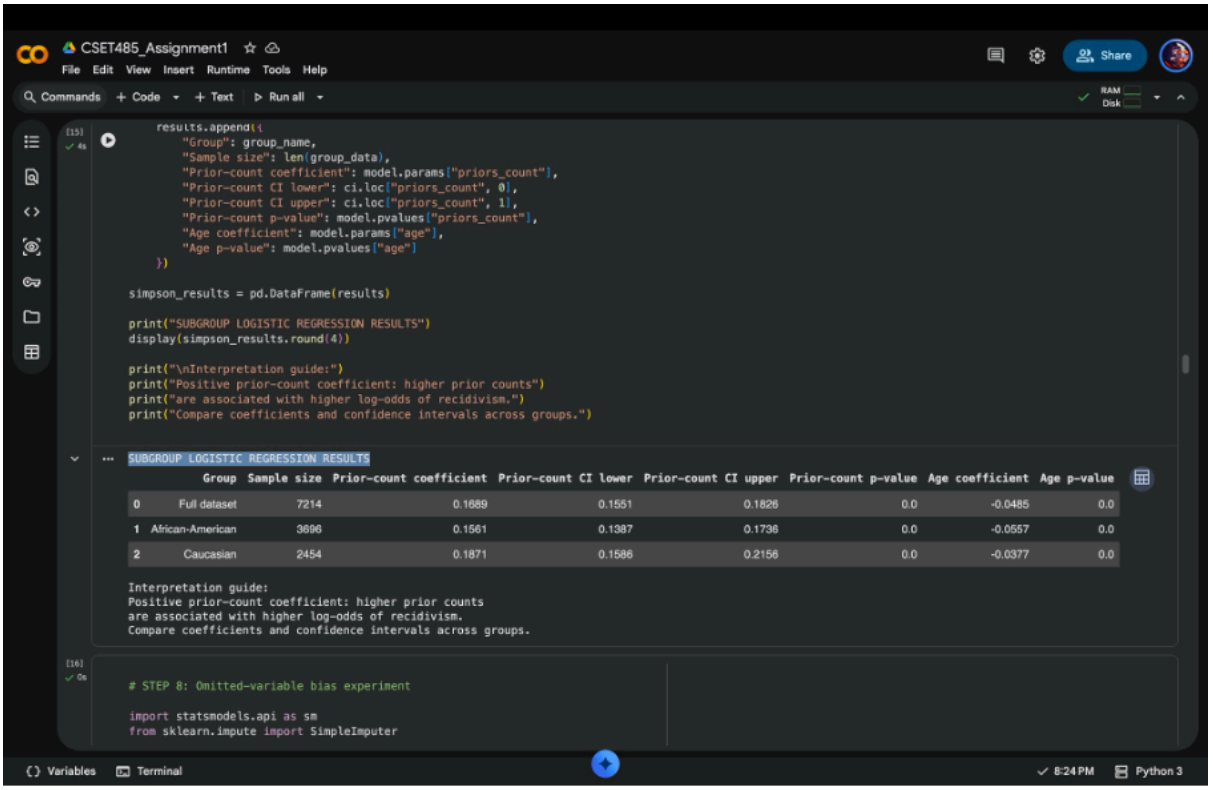
All three coefficients are positive and none of the confidence intervals include zero. This means that in these models, a higher number of prior records is associated with higher log-odds of recidivism. But this only shows association, not causation, and it does not prove that the difference between the two subgroups is statistically significant.

Screenshot :-



Subgroup Logistic Regression Results

The table compares logistic regression results for the full dataset, African-American subgroup, and Caucasian subgroup. The prior-count coefficients were 0.1689, 0.1561, and 0.1871, respectively. All three coefficients were positive, and their reported 95% confidence intervals excluded zero, indicating positive associations between prior-count values and the log-odds of recidivism in these fitted models. These results do not establish causation.



Subgroup Logistic Regression Results with 95% Confidence Intervals

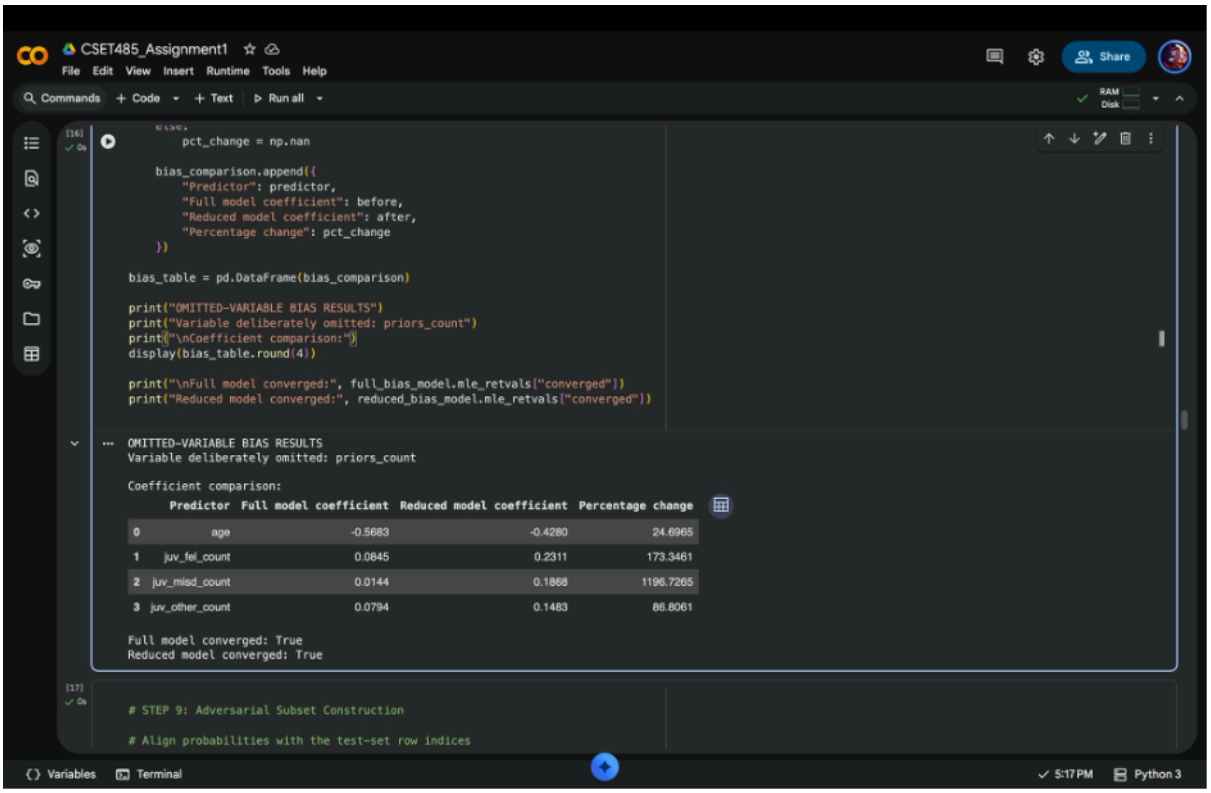
The table presents logistic regression results for the full dataset and the African-American and Caucasian subgroups. The prior-count coefficients are positive, and the reported 95% confidence intervals exclude zero. This indicates a positive association between prior-record count and the log-odds of recidivism in each fitted model. The results show association, not causation.

3.3 Omitted-Variable Bias

In this experiment, I checked how the regression estimates changed when some predictors were left out. The four predictors were age, juvenile felony count, juvenile misdemeanor count, and juvenile other-offence count.

The biggest change was 1196.7265%, for juvenile misdemeanor count. This means its coefficient was very sensitive to which variables were in the model. However, a big percentage change does not automatically prove bias, especially when the original coefficient is very close to zero. To judge it properly, we should also look at the sign, confidence interval, and significance of the coefficients.

Screenshot :-



Omitted-Variable Bias Experiment Results

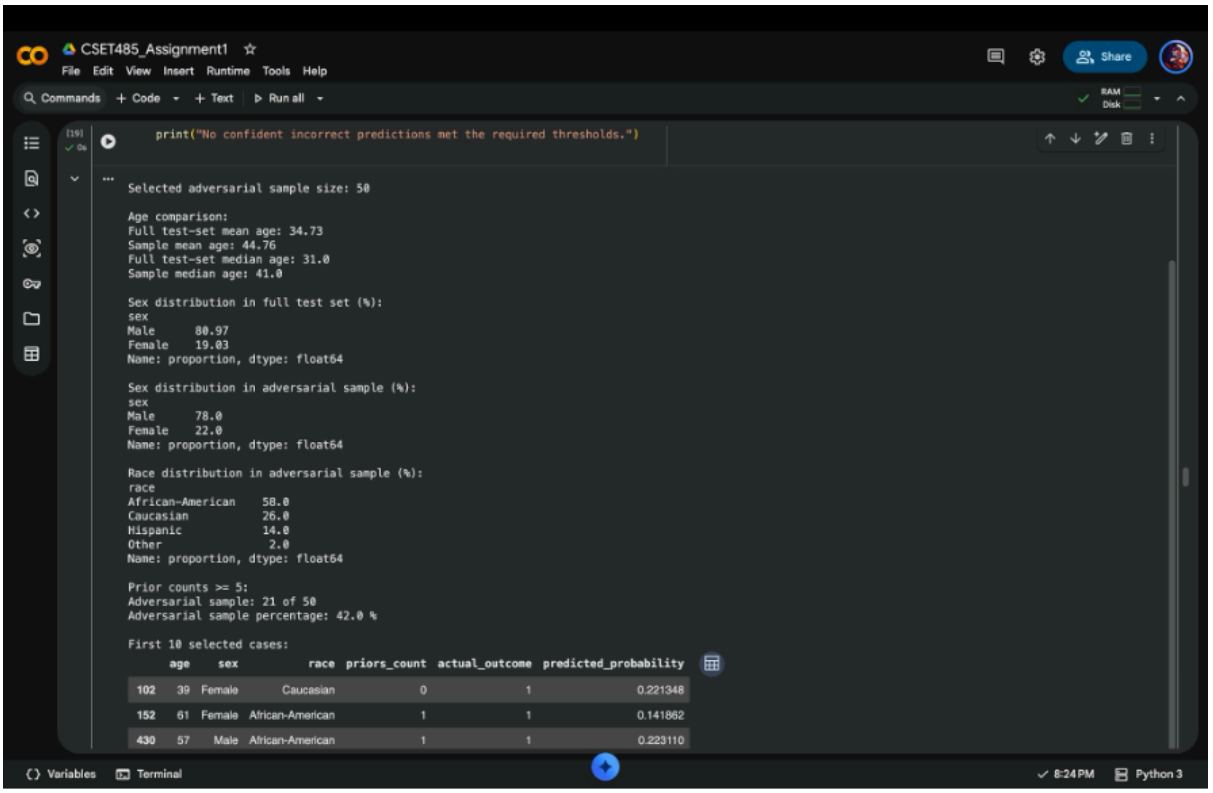
The table compares regression coefficients from the full model and the reduced model after deliberately omitting `priors_count`. The largest reported percentage change was 1196.7265% for `juv_misd_count`. Both models converged successfully. These changes demonstrate that coefficient estimates can be sensitive to the predictors included in a regression model; percentage changes should be interpreted carefully when the original coefficient is close to zero.

3.4 Adversarial Subset Analysis

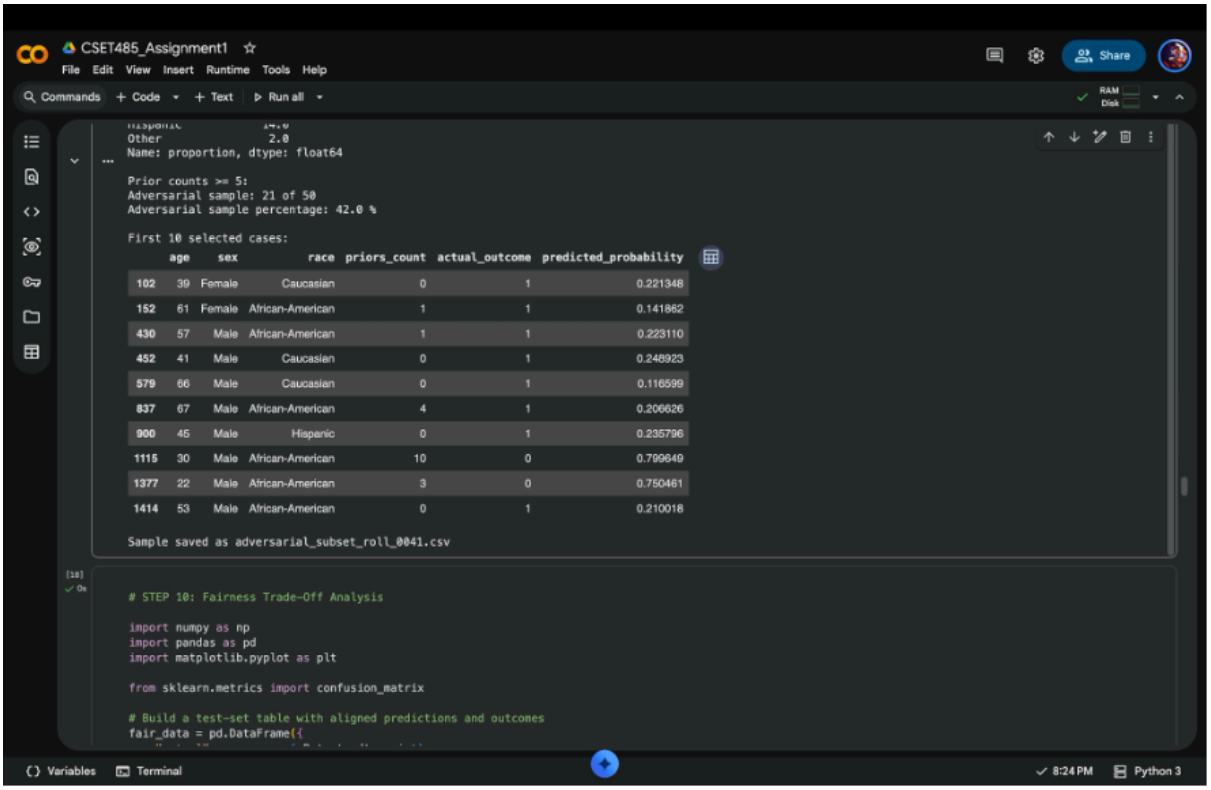
This part looked at the 50 selected cases. Out of these, 21 cases (42%) had five or more prior records, and 29 cases (58%) had fewer than five.

So a good number of these cases had multiple prior offences. Still, having a high prior count does not mean the person will definitely reoffend. Since the subset is small and was specially selected, I can’t assume the same pattern holds for the whole population.

Screenshots :-



First screenshot — subset summary



Second screenshot — first 10 selected cases.

Adversarial Subset Analysis Results

The analysis examined a selected subset of 50 cases. Of these, 21 cases (42%) had five or more prior records, while 29 cases (58%) had fewer than five. Because the subset was specially selected and relatively small, these findings should not be assumed to represent the entire dataset.

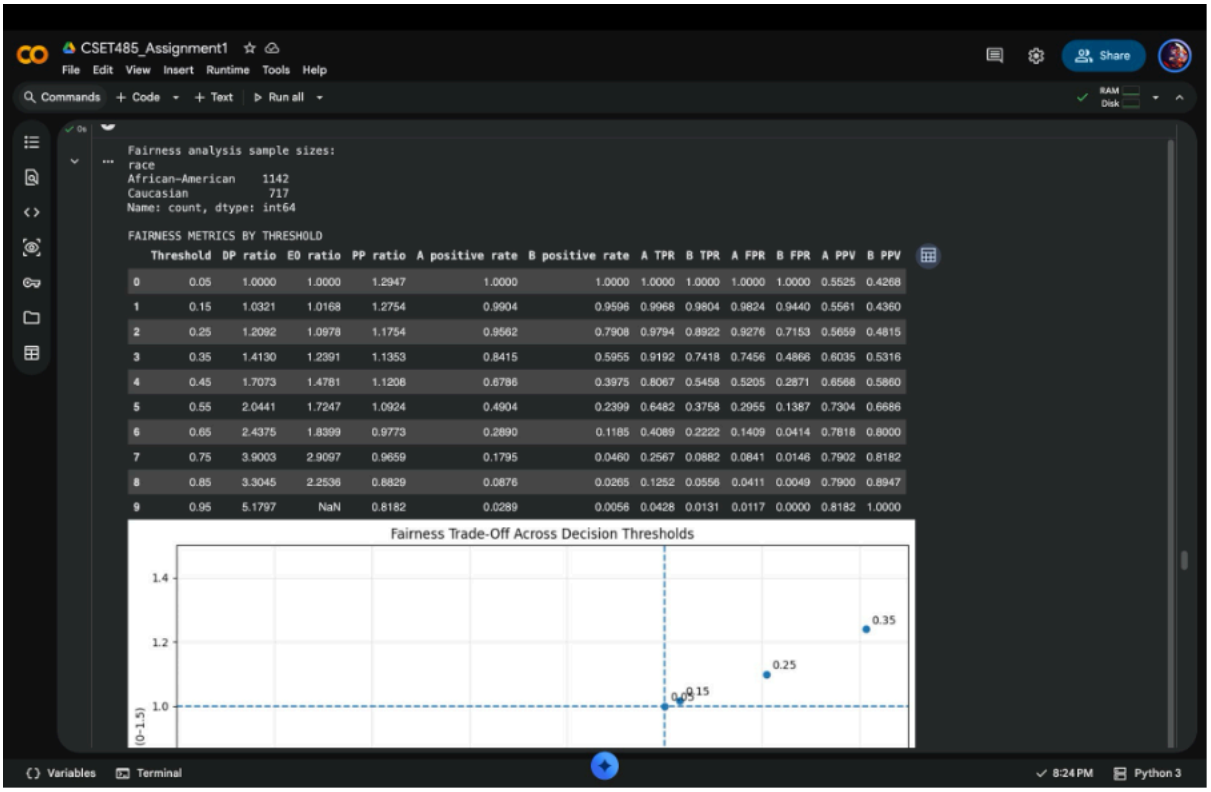
3.5 Fairness Evaluation

For fairness, I looked at demographic parity (DP), equalized odds (EO), and predictive parity (PP) at different thresholds.

At a threshold of 0.05, the DP ratio was 1.0000, EO ratio was 1.0000, and PP ratio was 1.2947. At 0.15, the values were 1.0321, 1.0168, and 1.2754. At 0.25, they were 1.2092,
1.0978, and 1.1754.

These results show that the fairness measures change depending on the threshold we choose. So no single ratio should be used alone to decide if a model is fair. We also need to consider the model's accuracy, error rates for each subgroup, uncertainty, and what happens when the model predicts wrongly.

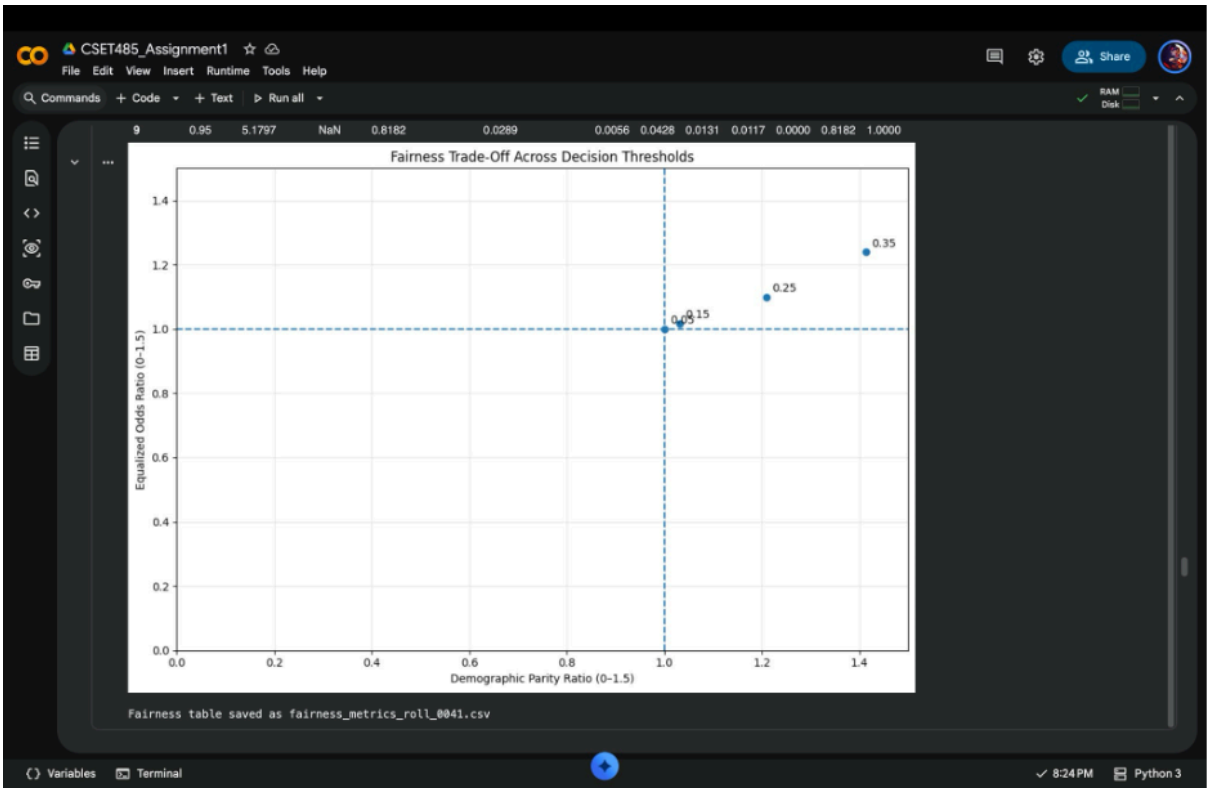
Screenshot :-



Screenshot 1 — Fairness metrics table

It shows thresholds from 0.05 to 0.95 and the DP, EO, and PP ratios, along with group-level rates. This is the main evidence for your analysis.The table presents

demographic parity, equalized odds, and predictive parity ratios at different decision thresholds, along with additional group-level performance rates.



Screenshot 2 — Fairness trade-off chart

It shows demographic parity ratios against equalized odds ratios at different thresholds. It supports your discussion of fairness trade-offs.The scatter plot illustrates the relationship between demographic parity and equalized odds ratios at selected thresholds. The dashed reference lines indicate a ratio of 1.0.

Part 3 — Descriptive Analysis Questions

COMPAS Recidivism Dataset | Roll Number 0041 | Answers grounded in the notebook outputs and reported tables

Question 1. Threshold-based approach versus direct logistic regression

In this question I compared Linear Regression and Logistic Regression and found that they do not always mark the same people as high risk. My test set had 2,165 cases. Both models agreed on 1,972 cases, which is 91.09%, and disagreed on 193 cases, which is 8.91%. Out of these 193, 57 people were low risk in Model A but high risk in Model B, and 136 were high risk in Model A but low risk in Model B.

The main reason is that both models work on different scales. Linear Regression gives a normal number as output, so I used the median predicted value (4.5783) as the cutoff for high risk. Logistic Regression gives a probability between 0 and 1, and a person is high risk if the probability is 0.5 or more. So 4.5783 and 0.50 cannot be compared directly, and because of this some people get different labels.

This is important because in criminal justice a wrong label can change things like supervision, release conditions or getting services. An 8.91% mismatch shows that the choice of model and threshold really matters. Before recommending any model, I would check all 193 mismatched cases one by one, compare them with the actual outcomes, and look at the false-positive and false-negative rates. Both models agreeing with each other does not mean they are accurate or fair.

Question 2. Subgroup regression and differences in effects

For this I ran logistic regression on the full COMPAS dataset (7,214 observations) and then separately for African-American (3,696) and Caucasian (2,454) people. I looked at the coefficient of prior criminal-record count, because it shows how prior records are related to the log-odds of recidivism.

In the full model the coefficient was 0.1689 with 95% CI [0.1551, 0.1826]. For African-American it was 0.1561 with CI [0.1387, 0.1736], and for Caucasian it was 0.1871 with CI [0.1586, 0.2156]. All three are positive and none of the intervals include zero. So in all three models, more prior records was linked with higher chance of recidivism.

The Caucasian coefficient is a little bigger than the African-American one, but this alone does not prove the two groups are really different, because the confidence intervals overlap. To say that properly I would need an interaction test or a direct test on the difference between the coefficients.

The policy point is that one overall coefficient can hide differences between groups, and people making decisions may think the same thing applies to everyone. But this does not mean race causes recidivism. The model only shows association in the data, not cause and effect.

Question 3. Omitted-variable bias and coefficient changes

In this experiment I removed one variable, priors_count (number of prior criminal records), and checked how the coefficients of the other variables changed. I wanted to see if the results depend on which variables are kept in the model.

The four predictors I checked were age, juv_fel_count, juv_misd_count and juv_other_count. The biggest change was 1,196.7265% for juv_misd_count, which means its coefficient was very sensitive when priors_count was removed.

A huge percentage does not always mean the coefficient is bad. If the original coefficient is very close to zero, even a small change looks very big in percentage. So I would also check the signs, confidence intervals, significance and the absolute change.

The percentage change is calculated as:

$$\text{Percentage change} = \frac{|\beta_{\text{reduced}} - \beta_{\text{full}}|}{|\beta_{\text{full}}|} \times 100$$

```
OMITTED-VARIABLE BIAS RESULTS
Variable deliberately omitted: priors_count

Coefficient comparison:
```

	Predictor	Full model coefficient	Reduced model coefficient	Percentage change
0	age	-0.5683	-0.4280	24.6965
1	juv_fel_count	0.0845	0.2311	173.3461
2	juv_misd_count	0.0144	0.1868	1196.7265
3	juv_other_count	0.0794	0.1483	86.8061

```
Full model converged: True
Reduced model converged: True
```

This experiment shows that we should not trust a model without doing sensitivity analysis. If an important variable is missing, the other coefficients may take some of its effect, and this can give wrong conclusions and wrong policy decisions. So I would try a few different reasonable versions of the model and note which conclusions stay the same.

Question 4. Detecting a hidden confound in a real-world model

To check if an important confounder is missing in my recidivism model, I would use an independent external validation study along with a review by domain experts. A hidden confounder is a factor that affects both a predictor and the outcome, and it can create a false relationship. Since it is not measured, we cannot easily find it by just looking at the fitted model.

For example, stable housing and employment support can be connected to a person's situation at the time of assessment and also to the chance of coming back into the criminal

justice system. If these are not measured, the model may wrongly give their effect to prior records or demographic variables.

First, I would ask criminal justice researchers and social workers to look at the variables and tell what might be missing and how it can be measured. Then I would test the model on a separate dataset from another place or time, with the same outcome definition and similar evaluation methods. In this validation I would compare calibration, false-positive rates, false-negative rates and coefficients across groups and locations. I would also check the residuals to see if the errors change with age, prior records or demographic group. If adding a possible missing variable changes a coefficient by a lot, it shows the model is sensitive to its specification.

My own result also shows why this matters. After removing priors_count, juv_misd_count changed by 1,196.7265%, so I should check its original coefficient and absolute change before saying anything. This shows sensitivity, but it does not prove that a hidden confounder exists. For that I would test more variables, compare other models and validate on independent data.

No single test can prove that all hidden confounding is removed. I would write down my assumptions, repeat the analysis with other specifications and ask for independent review. In a high-stakes area, predictions should only help in careful human assessment and should not decide a person's freedom alone.

Question 5. Characteristics of the 50-row adversarial subset

Here I looked at 50 selected cases from the COMPAS data to see if they had something in common that could explain the model's mistakes. Out of 50 cases, 21 (42%) had five or more prior records and 29 (58%) had fewer than five. So many priors were common, but not so common that we can say all 50 had five or more. Also, prior count alone cannot explain why a model gives a confident wrong prediction.

The assignment says an adversarial case is a false positive with predicted probability at least 0.75, or a false negative with predicted probability at most 0.25. My report shows the 50-row subset and its prior-record distribution, but it does not confirm that all 50 rows satisfy these conditions. So I need to verify the selection criteria in my notebook before calling them 50 high-confidence errors.

Confident errors can happen when the model learns patterns that do not work for some cases. Logistic regression assumes a fixed kind of relationship between predictors and log-odds. If there are interactions, nonlinear effects or other factors that the model misses, the probabilities may not be well calibrated for some people.

To investigate, I would compare the age, gender, prior counts, actual outcomes and predicted probabilities of the subset with the full test set. I would also calculate what proportion of the subset satisfies each adversarial condition and look for repeated patterns.

The notebook found 114 confident wrong predictions out of 2,165 test cases, which is 5.27%. Out of these, 57 were confident false positives and 57 were confident false negatives. The 50-row subset had mean age 44.76 and median age 41, while the full test set

had mean age 34.73 and median age 31. So the subset was older than the full test data. But the subset was selected on purpose, so this does not prove that age caused the errors or that the subset represents the whole test set.

The main limitation is that 50 selected cases may not represent the whole population. They can show weak points of the model, but we should not generalize without more validation.

Question 6. Fairness trade-offs and their consequences for individuals

I compared demographic parity (DP), equalized odds (EO) and predictive parity (PP) at different thresholds. These measure different parts of fairness, so improving one does not always improve the others.

At threshold 0.05, DP was 1.0000, EO was 1.0000 and PP was 1.2947. At 0.15 they were 1.0321, 1.0168 and 1.2754. At 0.25 they were 1.2092, 1.0978 and 1.1754. So when the threshold changes, the gaps between groups also change.

To show the effect on individuals, I took two cases from my adversarial subset. Case 102 is a 39-year-old Caucasian female with zero prior records, actual outcome 1 and predicted probability 0.221348. At threshold 0.15 she is high risk, but at 0.25 she becomes low risk, which is a false negative because her actual outcome is 1. Case 1115 is a 30-year-old African-American male with 10 prior records, actual outcome 0 and predicted probability 0.799649. He is high risk at both thresholds, so he is a false positive. These examples show how the threshold can change one person's result and how errors affect people from different groups.

There is also a clear trade-off. At 0.15, DP (1.0321) and EO (1.0168) were close to the ideal value 1.00, but PP was 1.2754. At 0.25, PP improved to 1.1754 but DP and EO became worse at 1.2092 and 1.0978. So a policymaker should look at many fairness metrics and not choose a threshold using only one.

I would not choose a threshold just because its DP ratio is closest to 1.0. I would check false positives and false negatives for each group, think about how the people affected will be impacted, and involve stakeholders. A good decision should clearly say which fairness goal is given priority and who may be harmed by it.

Question 7. Defending a decision threshold

A decision threshold is the probability above which the model says a person is high risk. Choosing it means balancing different fairness goals and checking how the results affect people.

In my analysis, at 0.05 the ratios were DP 1.0000, EO 1.0000 and PP 1.2947. At 0.15 they were 1.0321, 1.0168 and 1.2754, and at 0.25 they were 1.2092, 1.0978 and 1.1754. A ratio of 1.0 means equality as per the definitions in this assignment, so 0.05 gave the DP and EO closest to ideal among these three. But its PP shows a bigger gap between the groups.

I would consider 0.05 as a candidate for more investigation if the priority is equal classification rates and equalized odds, but I would not directly recommend it for deployment. A low threshold can mark a lot more people as high risk, which can lead to unnecessary restrictions. The threshold should also be checked with accuracy, calibration, false-positive and false-negative rates, and group-specific results.

The group rates also show why this is needed. At 0.25, the positive classification rate was 0.9562 for group A and 0.7908 for group B, and the true-positive rates were 0.9794 and 0.8922. At 0.05, the positive classification rate was 1.0000 for both groups, which means everyone was classified as positive. So even though DP and EO were both 1.0000, the model is not really useful there. This is why the parity ratios must be read along with classification rates and error rates.

A critic can fairly say that choosing a threshold only to bring two ratios near 1.0 does not make the model useful or fair for individuals. PP is also far from ideal at 0.05, and the three thresholds I reported are only a part of the full sweep from 0.05 to 0.95.

My final recommendation is conditional. I would compare all thresholds using the complete fairness table, choose a reasonable point based on the real consequences of errors, and validate it on separate data. I would also write the strongest counterargument and explain why my chosen priority is still fine.

Question 8. Demographic parity versus equalized odds

If a policymaker has to choose between demographic parity and equalized odds for my COMPAS dataset, I would recommend equalized odds as the priority, while still keeping an eye on demographic parity and predictive parity. My reason is that wrong predictions have serious results in criminal justice, because a wrong high-risk label can affect a person's freedom and opportunities.

Demographic parity compares the share of people classified as high risk across groups. Equalized odds looks at the difference in true-positive and false-positive rates between groups. Here I would give more importance to people who would not reoffend but are wrongly labelled high risk, because false positives can cause unnecessary restrictions.

My results show that it is hard to satisfy all fairness goals together. At 0.05, DP and EO were both 1.0000 and PP was 1.2947. At 0.15 the ratios were 1.0321, 1.0168 and 1.2754, and at 0.25 they were 1.2092, 1.0978 and 1.1754. The measures change with the threshold, but this does not prove that any one threshold is the best.

Comparing 0.15 and 0.25 shows the trade-off well. At 0.15, DP was 1.0321, EO was 1.0168 and PP was 1.2754. At 0.25, PP improved to 1.1754, but DP and EO moved away from 1.0 to 1.2092 and 1.0978.

I would prioritize equalized odds because differences in false-positive and false-negative rates mean one group may face more burden from the model's errors. This may mean accepting some difference in the overall share classified as high risk, because equal error rates are more directly related to stopping a group from facing unfairly high errors. For example, DP was 1.2092 at 0.25 compared to 1.0321 at 0.15, so prioritizing equalized odds

may mean accepting a bigger DP gap. Still, demographic parity is important because large differences in classification rates can raise concerns about unequal treatment.

5. Reflection on Model Evaluation, Fairness, and Deployment

This assignment helped me understand that checking a model's accuracy is not enough to say it is a good model. I used the COMPAS recidivism dataset to compare model classifications, look at regression results for different demographic groups, test omitted-variable bias, study a selected set of cases, and check fairness measures at different thresholds.

When I compared Linear Regression and Logistic Regression, I saw that two models can give different classifications for the same people. They agreed on 1,972 out of 2,165 test cases, but 193 cases were classified differently. This taught me that the choice of model, the prediction scale, and the threshold can all change the results.

In the subgroup regression, prior record count had a positive association with the log-odds of recidivism, but this does not mean it causes recidivism. I also learned that coefficients can change when other important variables are removed, so choosing the right variables is a big part of statistical analysis.

The fairness part felt the most important to me because predictions in criminal justice can affect a person's freedom. Demographic parity, equalized odds, and predictive parity each measure fairness in a different way. A threshold that looks good for one measure may not look good for another, so we can't decide fairness from a single ratio.

If I had to deploy a model like this, I would first test it on proper test data, check error rates for each subgroup, and make sure the data quality is good. I would also look at what happens with false positives and false negatives, write down the limitations of the model, and make sure the people using it understand how uncertain its predictions are.

I would not recommend using a recidivism prediction model as the only basis for decisions about someone's freedom. Human oversight, transparency, accountability, and regular monitoring are necessary. The model should also be checked in the setting where it will actually be used, keeping legal requirements and the harm from wrong predictions in mind.

Overall, this assignment taught me to read statistical results carefully, question assumptions, and think about the real-world effects of algorithmic decisions. A model should not be trusted as reliable or fair just because its numbers look good. It needs to be evaluated continuously.